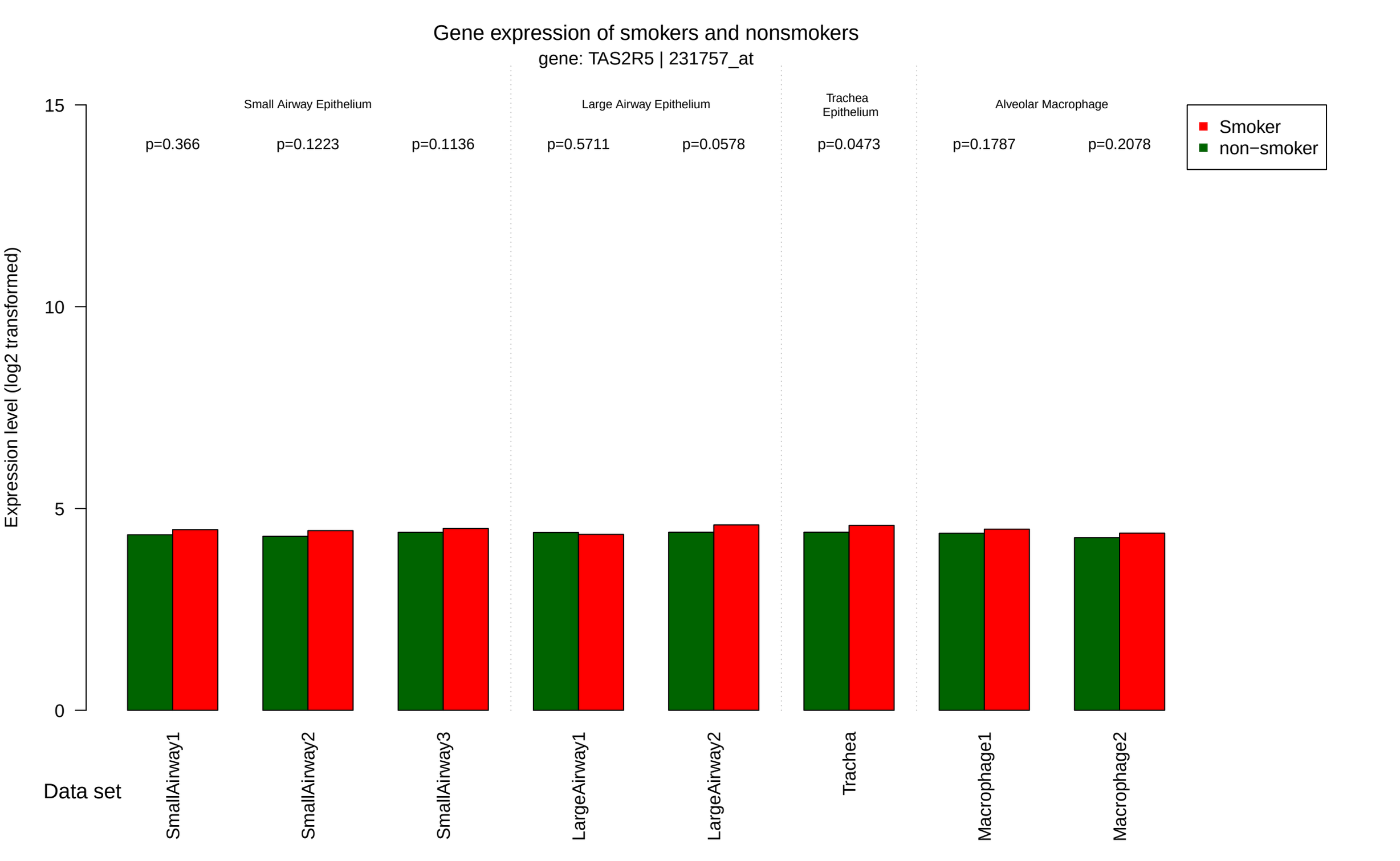
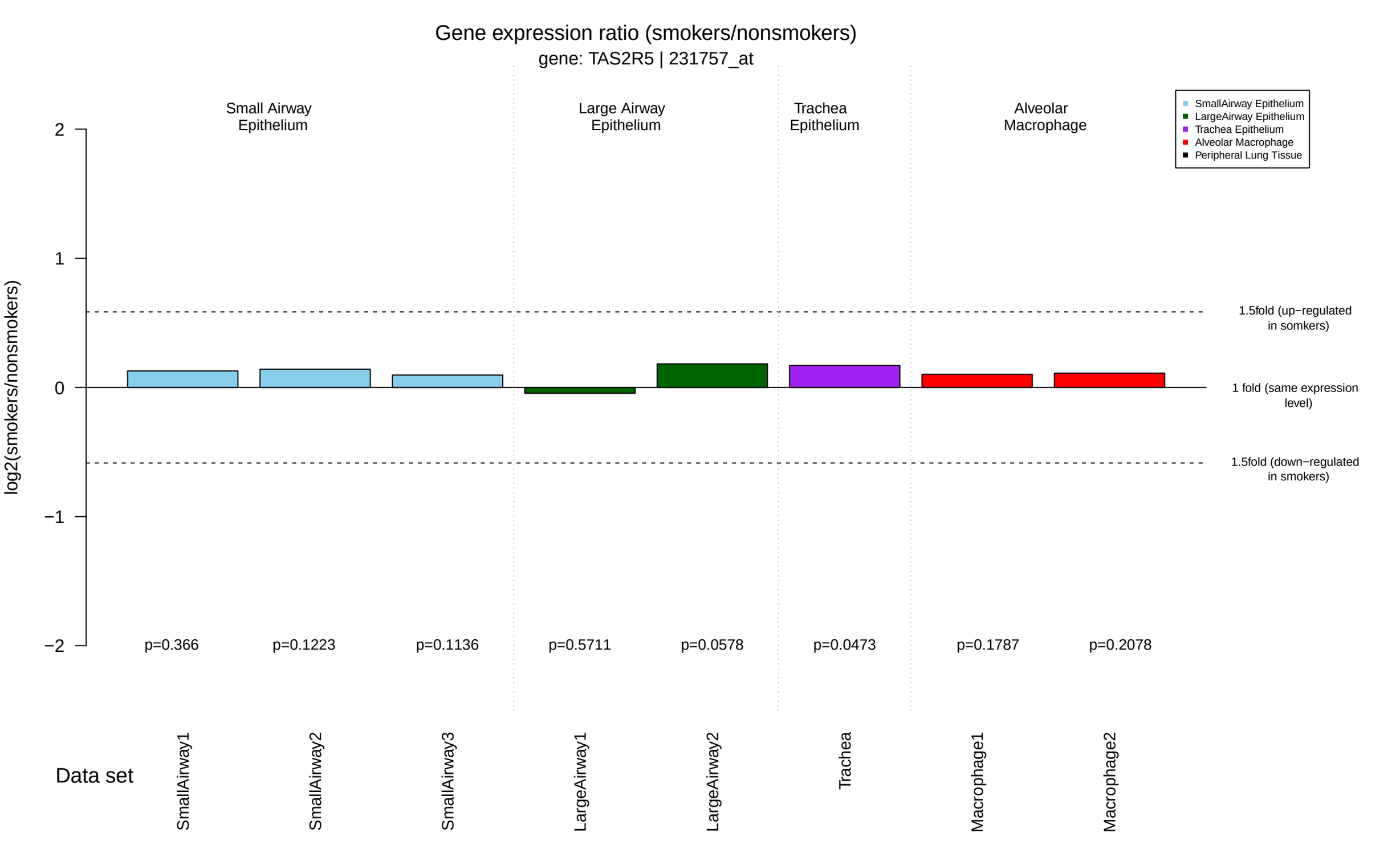
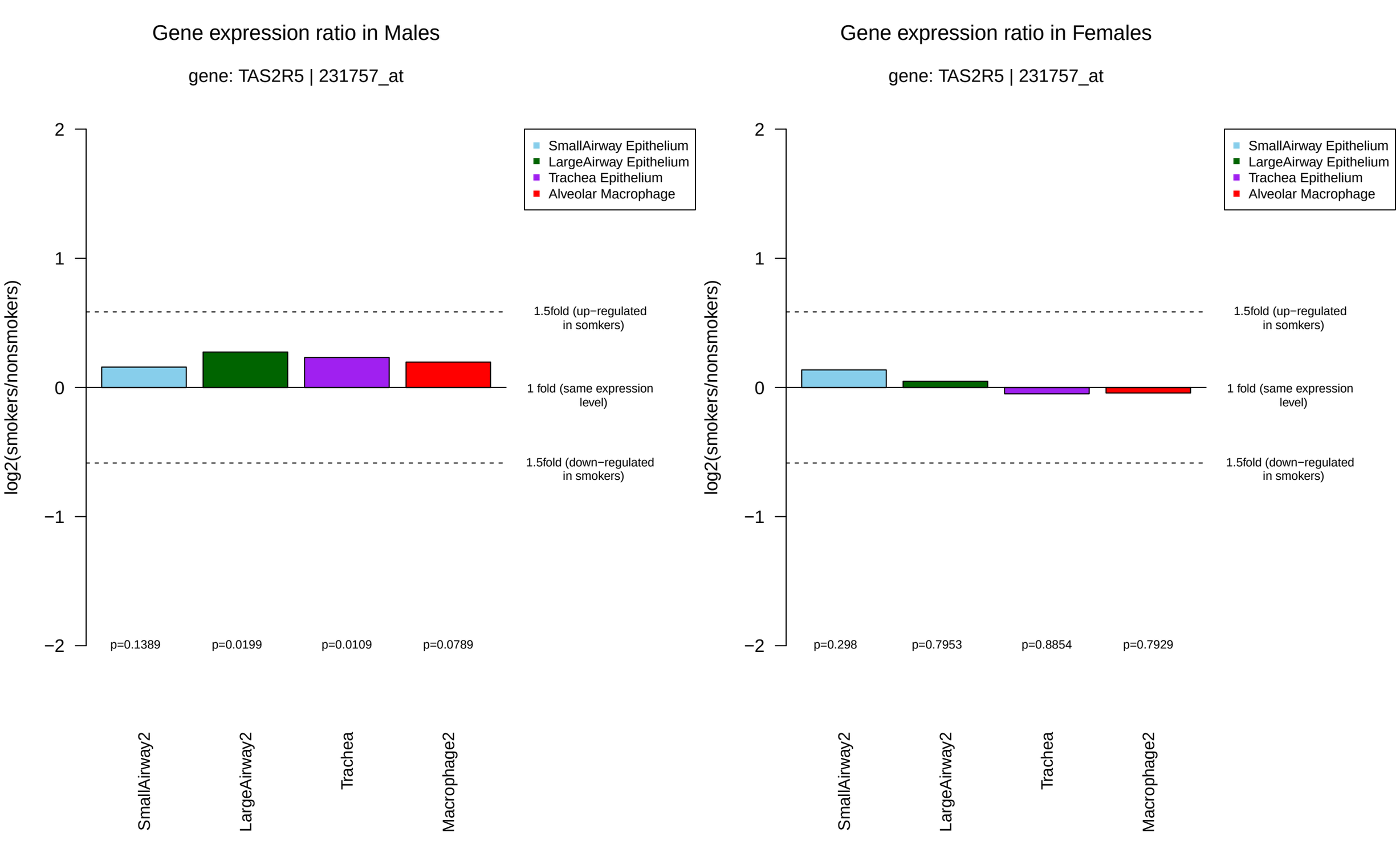
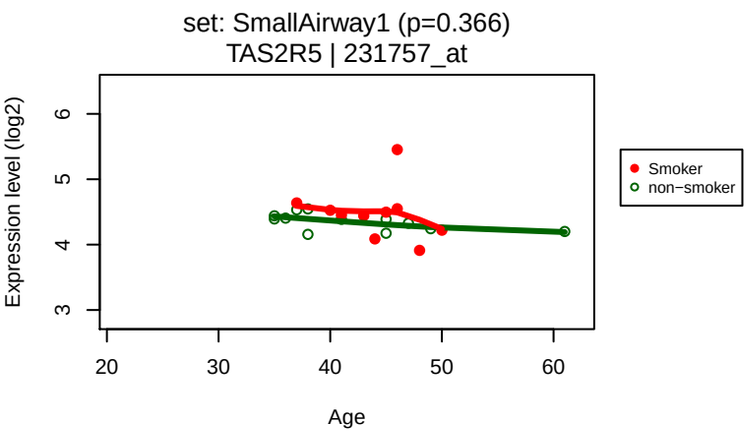
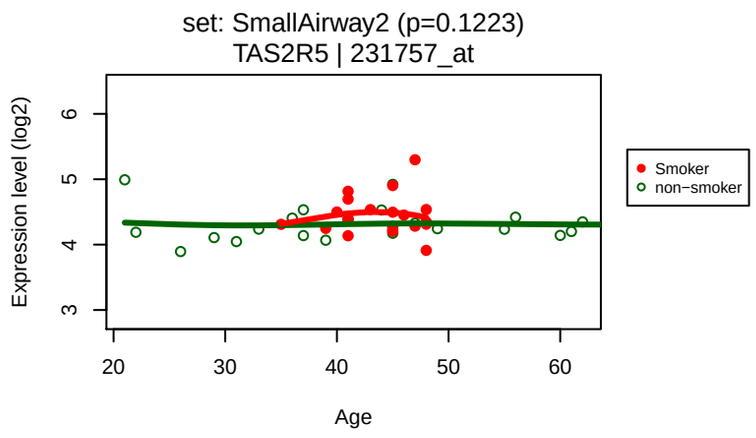
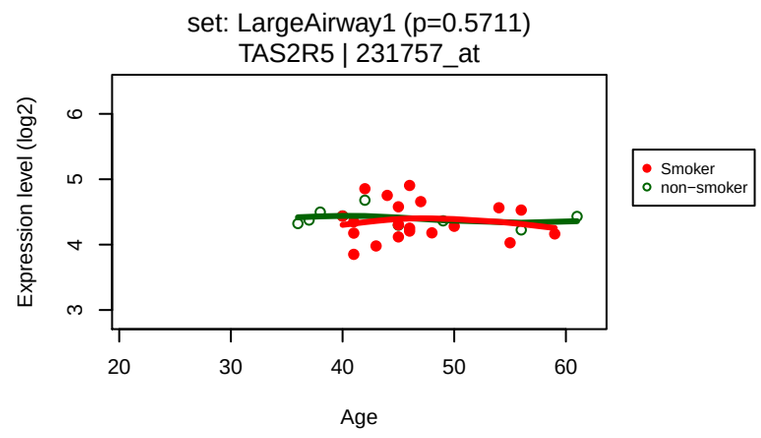

Gene expression of smokers and nonsmokers
gene: TAS2R5 | 231757_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.366
p=0.1223
p=0.1136
p=0.5711
p=0.0578
p=0.0473
p=0.1787
p=0.2078
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

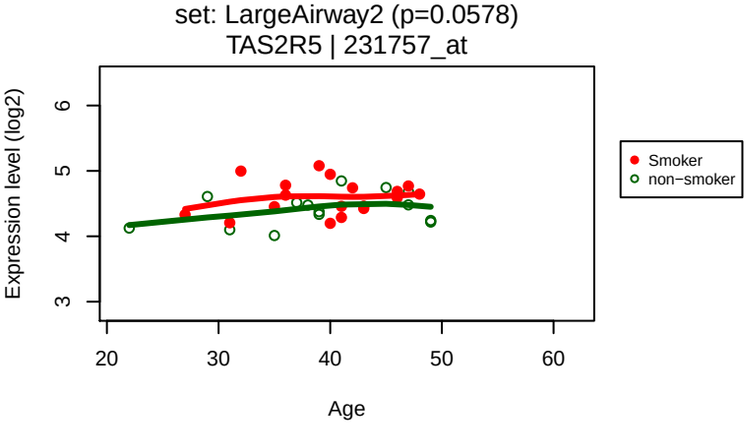

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R5 | 231757_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.366
p=0.1223
p=0.1136
p=0.5711
p=0.0578
p=0.0473
p=0.1787
p=0.2078
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

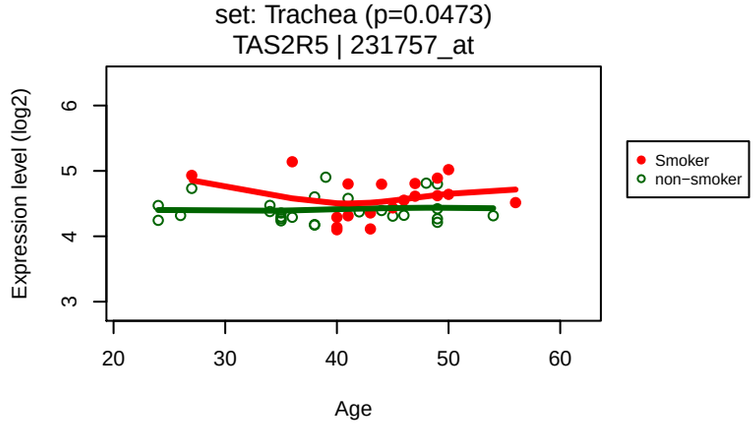

Gene expression ratio in Males
gene: TAS2R5 | 231757_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1389
p=0.0199
p=0.0109
p=0.0789
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R5 | 231757_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.298
p=0.7953
p=0.8854
p=0.7929
SmallAirway2
LargeAirway2
Trachea
Macrophage2

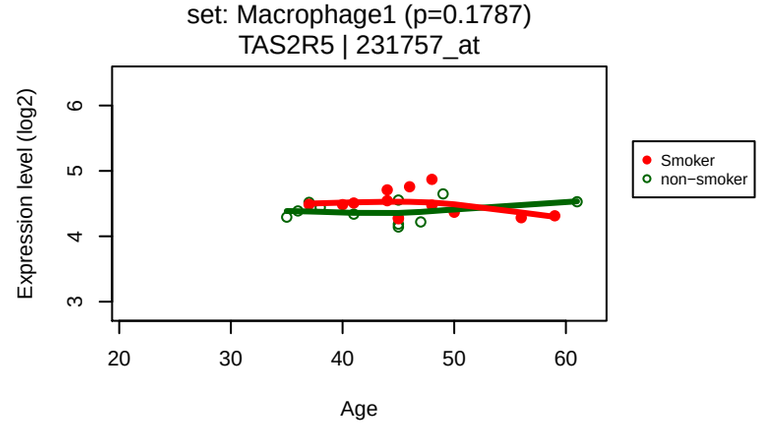

set: SmallAirway1 (p=0.366)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker

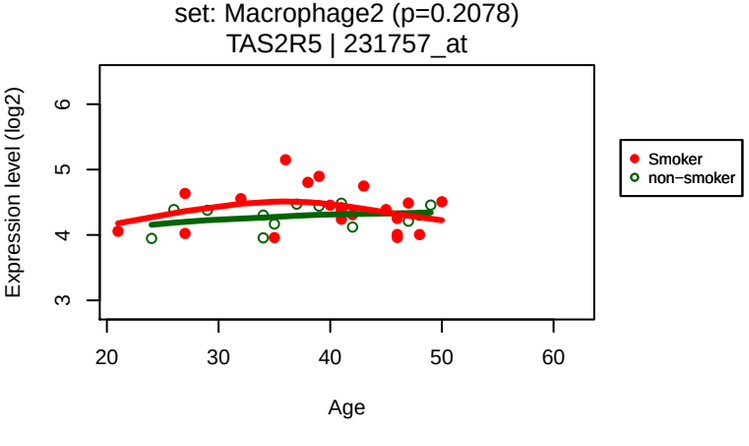

set: SmallAirway2 (p=0.1223)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker

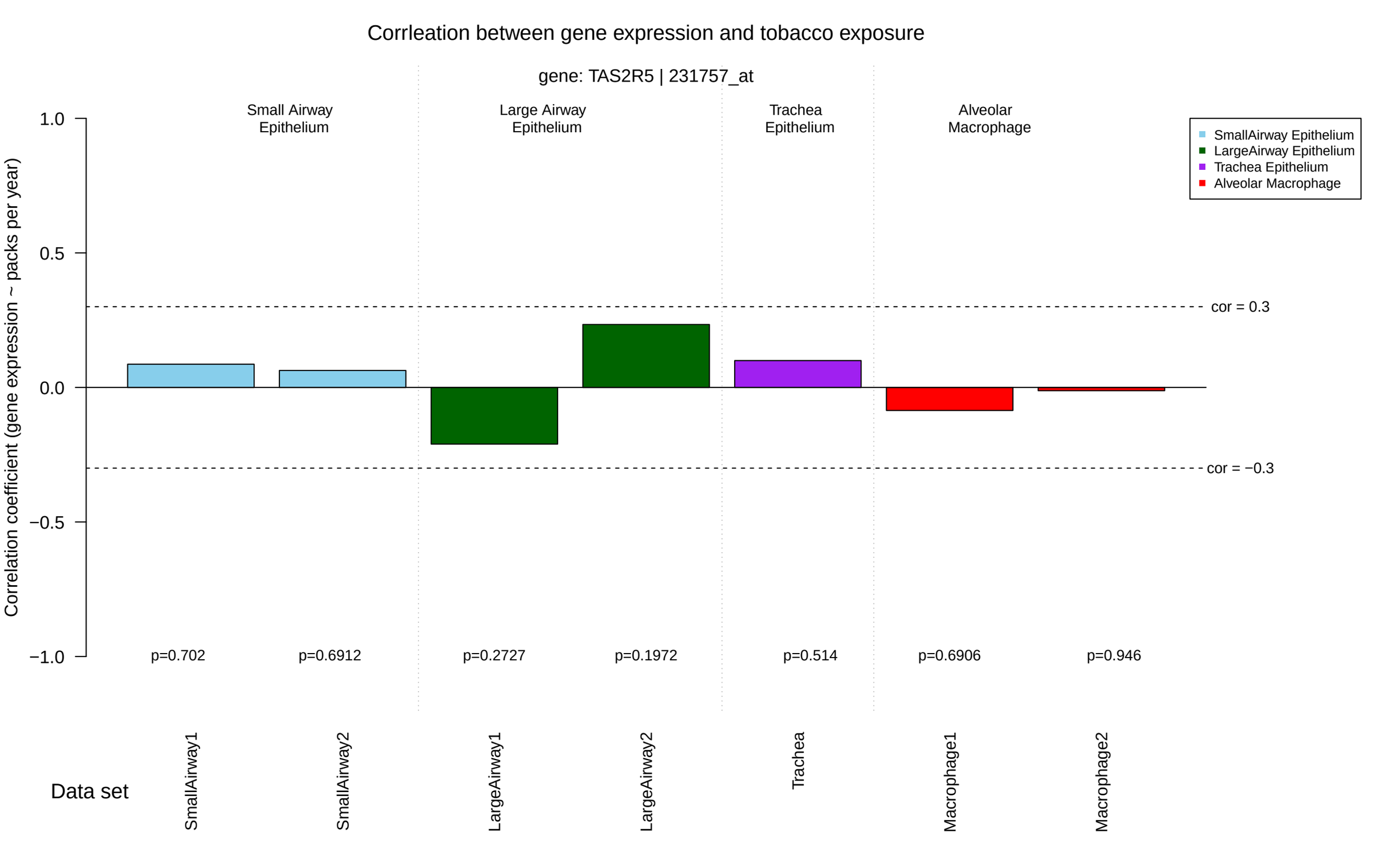

set: LargeAirway1 (p=0.5711)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0578)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0473)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1787)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2078)
TAS2R5 | 231757_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R5 | 231757_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.702
p=0.6912
p=0.2727
p=0.1972
p=0.514
p=0.6906
p=0.946
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2